

TUTORIAL-2

1. In Fig. 1, find the voltage across the 20Ω resistor using loop currents. If possible, choose the loop currents so that only one equation is necessary.

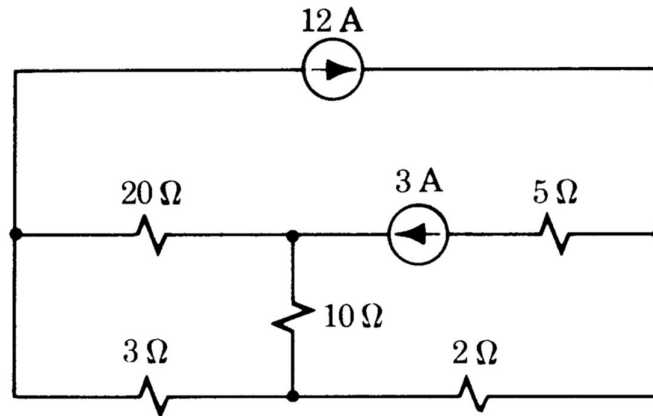


Fig. 1

2. Given the circuit of Fig. 2, consider various methods to calculate the current in the 5Ω resistance. Use the method that appears to require the least algebra.

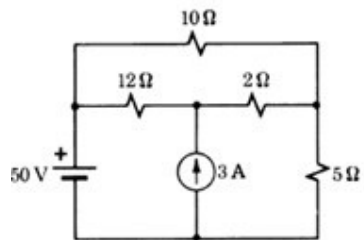


Fig. 2

3. Find the three currents in Fig. 3.

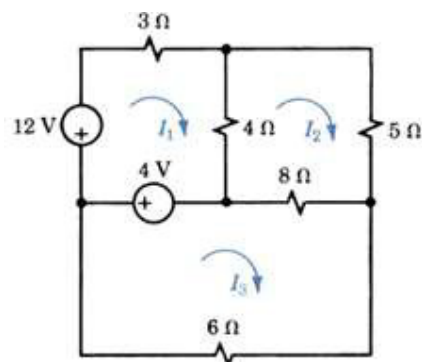


Fig. 3

4. Calculate the current gain i_o/i_s in the circuit of Fig. 4 if no current flows through the 10V source.

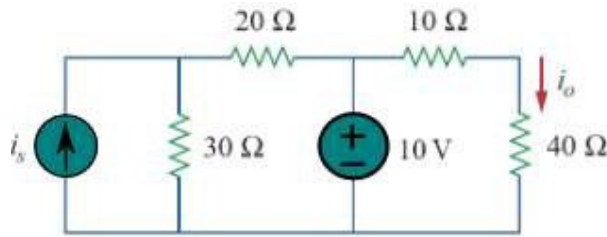


Fig. 4

5. Calculate the voltage gain v_o/v_s in the circuit of Fig. 5

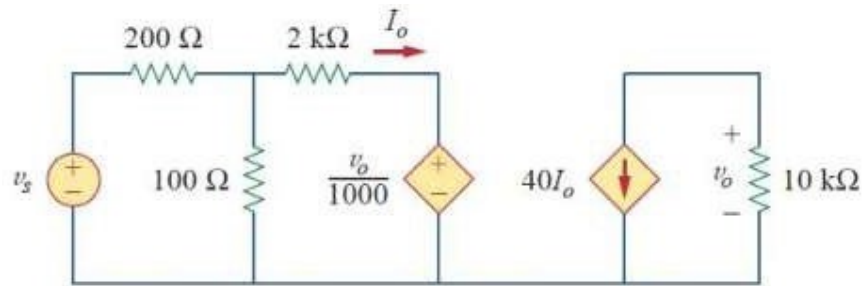


Fig. 5

6. (a) Attempt to apply node analysis to the circuit in Fig. 6. What prevents us from being able to write the node equations?
 (b) Using the fact that the current moving from node b into the 16 V source is the same as that moving from the 16 V source toward node a, and the fact that the relation between voltages at nodes a and b is known, write a single node equation encompassing both nodes and solve for v_a and v_b .

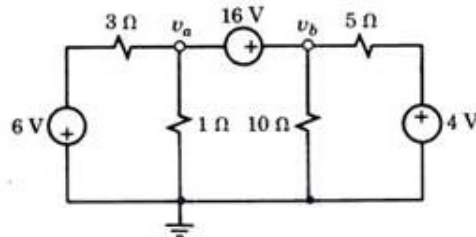


Fig. 6

7. A signal generator, consisting of a complicated network of nearly linear passive and active elements, supplies a variable load resistance R_L .
- Use Thevenin's theorem to replace the generator by a combination of V_G and R_G .
 - Prove the maximum power transfer theorem, which states that: "For maximum power transfer from a source to a load, the resistance of the load should be made equal to the Thevenin equivalent resistance of the source."

8. In an amplifier a transistor can be represented by the two-port model in Fig 7 where, $R_1=10k\Omega$, $R_2=75k\Omega$, $R_3=150k\Omega$, and $\beta=100$. (βI_1 is a controlled current source.)
- Simplify the model into a Thevenin equivalent and then predict the voltage amplification V_L/V_S as a function of R_L .
 - Calculate V_L/V_S for $R_L=20k\Omega$.

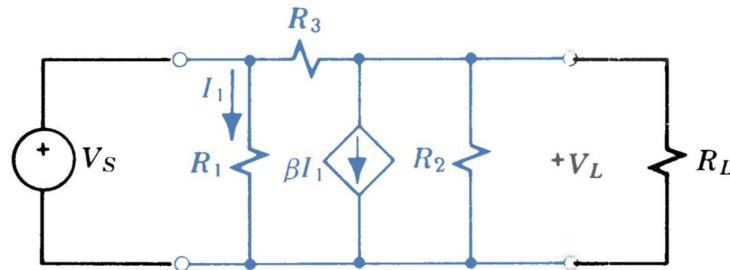


Fig. 7

9. Obtain the Thevenin and Norton equivalent at terminals a-b of the circuit shown in Fig. 8.

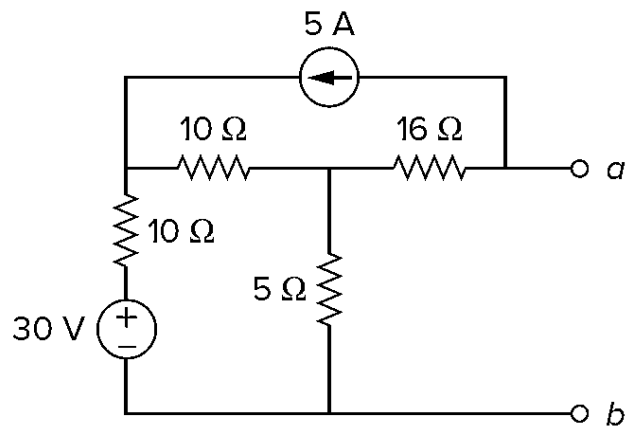


Fig. 8

10. Apply the superposition theorem to solve for the voltage V_0 in circuit shown in Fig.9

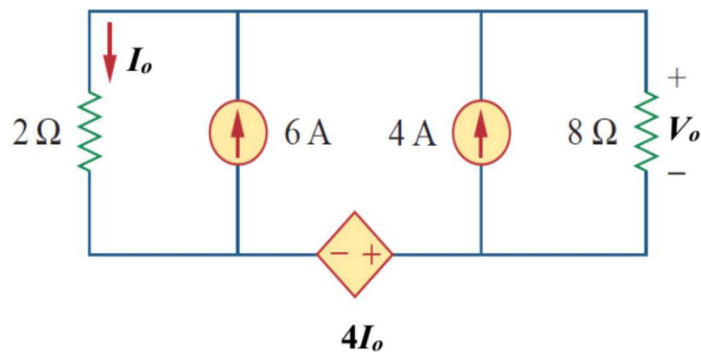


Fig. 9